



Department of Electrical and Electronics Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE - A

Course Code & Title: EE8591 Digital Signal Processing

Name of the Faculty member (s): Mr. D. Karthik Prabhu

Innovative Practice Description

- **Unit / Topic:** Unit I / Classifications of Systems
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO3
- **Activity Chosen:** Mind Map
- **Justification:**

Since Classification of Systems is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing mind map activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**

(A brief content about how the practice was followed)

A mind map is a graphical way to represent ideas and concepts. It is a visual thinking tool that helps structuring information, helping students to better analyze, comprehend, synthesize, recall and generate new ideas. It is an easy way to brainstorm thoughts organically without worrying about order and structure. It allows the student to visually structure the ideas which is used to help with analysis and recall.

The students were randomly called and they were asked to write the topic they had learnt in the classroom as a pictorial representation in the white board.

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the exam. In the Internal Assessment Test-I most of the students attended the questions under this topic and scored good marks.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO9	PO10	PO11	PSO1
CO1	3	3	3	3	3	3	3	3	3	3

(1 – Low

2 – Moderate

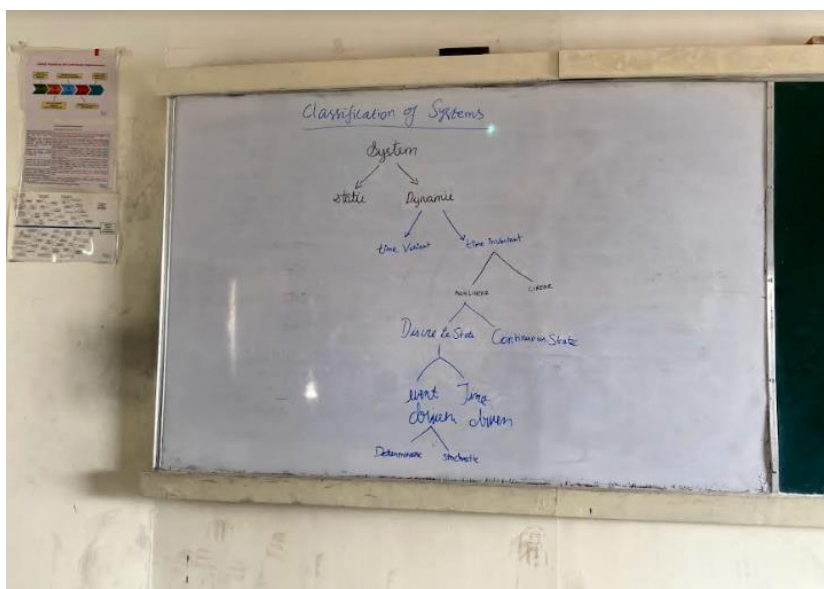
3 – High)



- PO / PSO mapped:

Innovative practice	PO10
	2
Justification for correlation	Due the Mind Map activity students written communication will be improved. So it is mapped strongly.

- Images / Screenshot of the practice:





- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students were asked me to conduct such activities for forth coming classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students gave oral feedback that now they got a clear idea about classification of systems, and they asked me to conduct more such activities like this.

- ❖ ***Challenges faced in implementation:***

I planned to conduct this activity within 10 minutes. But it took 15 minutes to finish this activity, since few students got hesitate to come to the board and write, I made those students to come forward. So it took 5 minutes extra time to finish this activity.

References:

- ❖ <https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/lecture-notes>
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
- ❖ Tarun Kumar Rawat, 'Digital Signal Processing', Oxford.

Signature of Faculty Member

HOD



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Course Code & Title: EE8591 Digital Signal Processing

Name of the Faculty member (s): Mr. D. Karthik Prabhu

Innovative Practice Description

- **Unit / Topic:** Unit II / Convolution
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO10
- **Activity Chosen:** One minute paper
- **Justification:**

Since Convolution is an important topic in concept wise and university exam point of view, I have chosen this topic and the reason for choosing One minute paper activity is while recollecting the concepts and hints learnt in the classroom, the students will be able to reproduce the concept in the exam.

- **Time Allotted for the Activity:** 03 Minutes
- **Details of the Implementation:**

(A brief content about how the practice was followed)

After teaching the concepts, I asked the students to think about the topic without writing anything for 2 minutes.

Total strength: 31

Reporter: Myself

At the end of the class (Last 4 minutes)

- ✓ I asked the students to think about what is convolution? How it is influence on the signal? For 2 minutes
- ✓ Then I asked the students to write as much as they can within a short period of time(1 minute)
- ✓ Finally, I collected the papers form each students (1 Minute)

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO8	PO10	PSO1
CO2	3	3	3	3	3	1	1	3

(1 – Low

2 – Moderate

3 – High)



- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Due the One-minute paper activity students written communication will be improved. So it is mapped moderately

- **Images / Screenshot of the practice:**



Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

Students were asked me to conduct such activities for forth coming classes.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The assessment of effectiveness of the activity was felt good when written most of the points by students.
- ✓ While conducting the activity, I understood that the students will be able to explain the operation of convolution on signals without any confusion.
- ✓ The success of the activity was evaluated by asking the question in Internal Assessment Test - II



❖ ***Challenges faced in implementation:***

- ✓ Time utilization for this activity
- ✓ Slow learners were not able reproduce the much points.

References:

- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
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Innovative Practice Description

- **Unit / Topic:** Unit III / Properties of DFT
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO12
- **Activity Chosen:** Zero Minute Speech

- **Justification:**

In general, the students won't study the properties of Fourier series and in the exam also they prefer to choose problems rather than choosing properties of Fourier series. But if they attend properties they can score full marks when compared to solving problems because there may be a chance to do mistake while solving problems. To change the mentality of the students and to make the students attend this topic in the exam, I have chosen this topic and conducted Zero Minute Speech Activity.

- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**

(A brief content about how the practice was followed)

The students were asked to recall the topic they had learnt in the classroom and they were randomly called to give speech for few minutes about the Properties of DFT.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO10	PSO1
CO1	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	3
Justification for correlation	Due the Zero Minute speech activity, students Oral communication will be improved. So it is mapped strongly.



- **Images / Screenshot of the practice:**



- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

Students were asked me to conduct such activities for forth coming classes.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

By conducting this activity the active learning skills of the students got increased and it was easy for them to recollect the points learnt in the classroom. They will also be able to reproduce the concepts in the IAT – II exam.

- ❖ **Challenges faced in implementation:**

I planned to conduct this activity within 5 minutes. But it took around 15 minutes to conduct this activity, since few students hesitate to come to the dais and share the concept. I changed their mentality and brought them to the stage to share the concept. Only very few students got chance to come and share their ideas since the time duration for conducting this activity is limited. Next time while conducting this activity, I thought of calling the students who have not participated in the previous activity (to give a chance for them).



References:

- ❖ Poorna Chandra S, Sasikala. B , Digital Signal Processing, Vijay Nicole/TMH,2013.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
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Innovative Practice Description

- **Unit / Topic:** Unit IV / Filter Design
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO21
- **Activity Chosen:** Class Poll
- **Justification:**
 - The steps involved for filter design will vary according to their types. The students must be able to understand which step is suitable for which filter. Hence the class poll is conducted for the chosen topic “Filter Design”.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 - The students sitting adjacent are made as a team of four for doing this activity.
 - The class poll questions have been shown to the class using the classroom projector.
 - The teams have provided their choices of the answers along with their reasons.
 - Finally, the correct and explanation have been provided by me.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1
CO1	3	3	3	3	3	3	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO9	PO10
	3	3
Justification for correlation	The students are working in a small group while doing this activity. Hence it is strongly mapped.	The students are communicating with each other in a team for getting the answer for the poll. Hence their communication skills are getting improved. So it is mapped strongly.



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The students have enjoyed the learning since the activity made them to the mood of small competition. At the same time, they have cleared their doubts about the concept. Most of the student told me to conduct such activity also in the future classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- ✓ The activity helped the students to gain insights about one another and build confidence in their own learning.
- ✓ It helped me to identify the gaps in my students comprehension about the topic.
- ✓ The activity is incredibly useful in avoiding the monotony of a lesson by breaking up the content flow.

- ❖ ***Challenges faced in implementation:***

The class poll can be created with so many online tools, so that each student can participate in the poll. But the lack of Computer / Laptop / Mobile and Internet connectivity not allowed me to conduct using those tools. Hence next time, I will plan this activity in laboratory.

References:

- ❖ Poorna Chandra S, Sasikala. B , Digital Signal Processing, Vijay Nicole/TMH,2013.
- ❖ John G. Proakis and D.G. Manolakis, 'Digital Signal Processing Principles, Algorithms and Applications', Pearson Education, New Delhi, PHI. 2003.
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Innovative Practice Description

- **Unit / Topic:** Unit V / Architecture of Digital Signal Processors
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO22
- **Activity Chosen:** Flipped Classroom

Justification:

Flipped Classroom is a pedagogical approach in which the conventional notion of classroom based learning is inverted, so that students are introduced to the learning material before class, with classroom time then being used to deepen understanding through discussion with peers and problem solving activities.

- **Time Allocation of the Activity:** 15 minutes

- **Details of the Implementation:**

The students were asked to recall the topic they had learnt in the classroom and they were randomly called to give speech for few minutes about the Architecture of Digital Signal Processors.

Due to this activity, the students were able to recall the topics they have learnt in the classroom and they were also able to reproduce it in the internal assessment test.

- **CO – PO / PSO mapping:**

Course Outcome	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1
CO3	3	3	3	3	3	3	3	3	3

- **PO / PSO mapped:**

Innovative practice	PO10
	3
Justification for correlation	Due the Mind Map activity students oral communication will be improved. So it is mapped strongly.



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

Students were asked me to conduct such activities for forth coming classes.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

By conducting this activity the active learning skills of the students got increased and it was easy for them to recollect the points learnt in the classroom. They will also be able to reproduce the concepts in the exam. The students gave oral feedback that now they got a clear idea about classification of digital signal processors architecture, and they asked me to conduct more such activities like this.

- ❖ ***Challenges faced in implementation:***

I planned to conduct this activity within 5 minutes. But it took around 15 minutes to conduct this activity, since few students hesitate to come to the dias and share the concept. I changed their mentality and brought them to the stage to share the concept. Only very few students got chance to come and share their ideas since the time duration for conducting this activity is limited. Next time while conducting this activity, I thought of calling the students who have not participated in the previous activity (to give a chance for them).

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